

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Supplementary Winter Examination: Dec-2019

Course: B. Tech in -Civil Engineering

Sem : IV

Subject: Numerical Methods in Engineering (BTCVE404A)

Max Marks: 60

Date:02/12/2019

Duration:- 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately Assume it and should mention it clearly.

Q.1A) $x + 2y + 3z = 5$

6x2

$$2x + 8y + 22z = 6$$

$$3x + 22y + 82z = -10$$

Solve the equation by using Cholesky method.

B) $83x + 11y - 4z = 95$

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71$$

Solve the equation by using Gauss Seidel method.

Q.2A) If $dy/dx = x+y^2$ and $y=1$ of $x=0$, find an approximate value of y at $x=0.2$ By Euler's modified method ($h=0.1$)

6x2

B) $x^2 + 2x - 0.5 = 0$ find the root upto 3 decimal places. Use Secant method

Q.3A) Find $f(42)$ for the following data

6x2

X	20	25	30	35	40	45
F(X)	354	332	291	260	231	204

Use Newton's backward difference method.

B) Find the population increase between 1946 to 1948. Use newton's forward difference method

year	1911	1921	1931	1941	1951	1961
Population (1000)	12	15	20	27	39	52

Q.4A) Evaluate the following integral using Simpson's 1/3 rule'.n=2.

6x2

1) $\int_{-1}^1 e^x dx$ 2) $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx$

B) Explain trapezoidal rule, Simpsons 1/3 rule, and Simpsons 3/8th rule.

Q.5A) Explain Mean, Median, Mode, and Deviation.

6x2

(B) Fit the straight line to the following set of the data

X	1	2	3	4	5
Y	3	4	5	6	8

6x2

Q.6A) Write the algorithm for Euler's method

B) Write the algorithm for Newton Raphson Method

Paper End